Q02: Is Next-Token Generation Intelligence? How Did Intelligence Emerge from Deep Learning?

DLS 2026 — Draft module

October 2026

Learning objectives

Students will distinguish an operational performance claim from a claim about intelligence; identify what scaling results do and do not establish; and design an evaluation that tests a stated capability rather than relying on a label.

1 Start by unbundling the question

"Intelligence" is not one measured quantity. It can mean successful task performance, flexible generalization, planning, learning, social understanding, consciousness, or some combination. A useful course answer therefore begins by asking: *which capability, in which environment, under which evidence standard?* Turing's 1950 proposal redirected a vague metaphysical question toward observable conversational performance [4]; it did not settle every later question about understanding, agency, or consciousness.

Next-token generation is a training objective: estimate $p(x_{t+1} \mid x_{\leq t})$. The objective alone neither proves nor rules out intelligence. A system can perform impressively on some tasks while failing brittle variants, relying on training-data regularities, or lacking abilities that a student associates with intelligence. The scientifically useful move is to replace a global label with claims that can fail.

2 What we can observe

Language models can produce useful behavior on many tasks: answer questions, transform text, write code, and adapt their output to demonstrations placed in context. Brown et al. documented few-shot task performance in large language models [2]. Such observations justify careful claims about measured behavior on stated tasks. They do not by themselves identify the internal algorithm, establish robust out-of-distribution generalization, or show that the system has human-like goals or experience.

A capability claim needs a task distribution, a scoring rule, comparison baselines, and a failure analysis. For example, "the model can reason" is too broad. "On held-out arithmetic word problems with adversarially varied wording, the model reaches X% exact accuracy and its verifier-checked intermediate program succeeds Y% of the time" is an empirical claim another group can test.

3 Why capabilities may change with scale

As data, compute, and model capacity change, measured performance can improve. Some reports describe sharp threshold-like gains on particular benchmarks [5]. There are at least three nonexclusive interpretations: an underlying computation becomes available only after sufficient capacity/training; a continuous capability crosses a task threshold; or the evaluation metric and prompting procedure make a smooth gain look abrupt. Schaeffer et al. show that metric choice can create apparent discontinuities [3]. "Emergence" is consequently a description to investigate, not a causal explanation.

Deep learning does not add a single "intelligence switch." Optimization can create representations and procedures that help predict structured data. Larger or better-trained models may use those representations more effectively in context. But how reliably that transfers depends on the data, architecture, post-training, prompt, tools, and evaluation conditions.

4 Three bad inferences to avoid

1. **Fluency implies understanding.** A coherent explanation can be false or copied from a familiar pattern. Check it against a new problem and a verifier.

2. **One benchmark implies general intelligence.** A benchmark measures a constructed slice of behavior and can suffer contamination, shortcut exploitation, or saturation.

3. **Failure implies no capability.** A failure may reflect a prompt, interface, tool restriction, or distribution shift. Diagnose before drawing a global conclusion.

These cautions do not make the systems uninteresting; they make the claims testable. They also connect to critiques that warn against treating text-generation behavior as a complete account of language understanding or social competence [1].

5 A small operational exercise

The accompanying code runs a deterministic rule-learning task. A rule is shown through examples, then evaluated on held-out combinations. It compares exact generalization with a deliberately weak memorization baseline. The exercise does *not* measure intelligence. It demonstrates a methodological point: define a capability, hold out the cases that distinguish rule use from lookup, and inspect errors before assigning a large label.

6 Takeaway

Whether to call a next-token system "intelligent" is partly a conceptual choice, but its component claims are empirical. Describe the behavior, specify the distribution and metric, test alternatives, and report uncertainty. Scaling studies motivate hypotheses about changing capability; they do not eliminate the need for careful evaluation.

References

[1] Emily M. Bender et al. On the dangers of stochastic parrots: Can language models be too big? *Proceedings of FAccT*, 2021.

[2] Tom B. Brown et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 2020.

[3] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? *arXiv preprint arXiv:2304.15004*, 2023.

[4] Alan M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950.

[5] Jason Wei et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.